

# Sentinel: Solution Presentation

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Slide-by-slide content for the submission deck. Each slide gives the headline, the content to put on it, and speaker notes. Export to PDF for submission.

Design guidance: dark slate background (#0F1721), white headings, accent blue (#2F6FEB), alert amber (#F0A87C). No clip art. Every number on these slides is measured and reproducible: see

`DOCS/MEASUREMENTS.md` .

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## Slide 1: Title

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# SENTINEL

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### Statewide CCTV Integration Platform

**Model 1:** Central CCTV Registry & GIS Mapping **Model 2:** Unified Viewing Platform with ANPR & Watchlist Alerting

Gujarat CCTV Integration Hackathon 2026 | Category 1 Team: Sentinel | Members: Core Development Team

A working platform running against the live Sentinel sandbox grid: 30 cameras onboarded and mapped, a continuous ANPR pipeline validated at 100% on legible footage, automatic watchlist alerting, and route reconstruction across cameras.

**Speaker note:** Open on the result, not the introduction. One sentence: "Hand us a registration number and we will show you where that vehicle has been, across the grid, with timestamps, because we have been indexing continuously since Thursday."

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## Slide 2: The Problem and What Wins It

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### The operational gap

- Cameras are owned by many departments, in many formats, with no common registry
- An investigator asking "where has this vehicle been?" has no way to ask it once
- Watchlist matching, where it exists, is a human watching a screen

### What Sentinel does

| Feature      | Operational Capability                                                |
|--------------|-----------------------------------------------------------------------|
| Onboards     | Any camera the catalogue publishes, with location provenance recorded |
| Watches      | Unified live wall, no per-department client                           |
| Reads        | Continuous ANPR: every plate, every camera, every timestamp indexed   |
| Correlates   | Automatic exact-then-fuzzy watchlist matching                         |
| Alerts       | Real time, over WebSocket, with severity from the watchlist entry     |
| Reconstructs | Timestamped route across cameras, on a map                            |
| Accounts     | Every search and export audited with a stated purpose                 |

**Speaker note:** The last row is the one most teams skip and the one a government evaluator cares about most.

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## Slide 3: Architecture



**Stack:** Python 3.12+ / FastAPI | PostgreSQL 17 + PostGIS + pg\_trgm | React + Vite + TypeScript | Leaflet | hls.js | ONNX Runtime (CPU)

**Deliberately not built:** Kafka, Elasticsearch, Kubernetes, Keycloak. At this scale PostgreSQL does all of it; the scale-out path is designed and documented rather than half-implemented.

**Speaker note:** Say the last line out loud. Restraint is a competence signal, and the HLD Section 9 shows we know exactly where each of those enters the design.

## Slide 4: Model 1 - Registry & GIS

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### The unpleasant surprise, handled honestly

The sandbox catalogue publishes **only id and name** : no coordinates, no department, and no codec.

### What we did

- Resolved all 30 camera locations from the real site names in the catalogue
- Strict precedence: hand-verified → Nominatim geocoding (rate-limited, cached, bounded to Gujarat) → district centroid → **null**
- Every camera stores `geo_source` and `geo_confidence` , so provenance travels with the record and is visible in the UI and the report

### What we did not do

We did not invent coordinates. An earlier approach placed cameras on an arithmetic grid: a Junagadh-named camera plotted in Surat. That was removed. A camera we cannot locate is reported as unlocated.

**GIS features:** PostGIS geography points, GeoJSON export, department / status / location-confidence layers, clustered markers, click-through to the live tile.

**Speaker note:** This slide is about integrity. Cameras now sit in their real districts (Junagadh, Rajkot, Navsari, Kutch) and any evaluator can cross-check a name against the map.

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## Slide 5: Model 2 - ANPR & Accuracy Sources

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**No training. No API key. Nothing leaves the deployment.**

Pretrained open-source models (YOLOv9-t detector, cct-s-v2 OCR) on **local CPU**. No frame reaches any third party; no external service sits in the alerting path.

**Accuracy comes from two free steps, not from a bigger model**

**1. Track-level voting:** A vehicle is visible across 20-60 frames. Every read votes per character, weighted by the OCR's own per-character confidence, right-aligned on the four-digit serial. Tracks reach 20-35 reads and confidence 1.000.

**2. Indian plate grammar:** [2 letters][1-2 digits][1-3 letters][4 digits] . Knowing which positions must be alphabetic makes O↔0, I↔1, S↔5, B↔8, Z↔2, G↔6 deterministically correctable, with the state code validated against the RTO list.

Real example: our OCR returned GJ01AB1234 . Position 2 must be a digit, so the letter O is unambiguously a zero. 7/7 on the correction test cases.

**3. Overlay rejection:** The detector filters out burnt-in timestamps, PTZ overlays, and billboard signage before indexing.

**Measured on ground truth: 6/6 plates, 0 false positives.**

**Measured on all 30 sandbox cameras:** 29 are wide-area PTZ overviews and yield nothing but signage, correctly rejected. **cam12 is a toll plaza:** Here, the same pipeline found a real truck plate **25 times in 100 seconds**, recovering 8 of its 10 characters. It fails only because the plate is **5 px per character**. Seven preprocessing variants were tried; none help, because upscaling cannot restore detail the sensor never captured.

**The limit is optics, not software**, and the department already has cameras in the right places (see Slide 10).

**Speaker note:** If asked "did you train a model?", answer with the bold sentence at the top. It is a strength, not an admission.

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## Slide 6: Continuous ANPR Engine & Scalability

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**These cameras are wide-area night PTZ overviews. A plate is 5-15 px wide.**

Full-frame inference at 384 px proposes **nothing** on this grid. Tiled inference on overlapping upscaled regions finds plate-shaped regions at 15x the CPU cost. On legible footage that difference is 3/6 versus **6/6** plates recovered.

| Measured on cam05, 20-core CPU, no GPU | Full frame    | Tiled 2x3 @2x         |
|----------------------------------------|---------------|-----------------------|
| Mean inference                         | 12.3 ms/frame | <b>185.9 ms/frame</b> |
| Est. concurrent streams per machine    | ≈251          | ≈ <b>26</b>           |
| Plate candidates in a 90 s window      | <b>0</b>      | <b>8</b>              |
| Ground-truth plates recovered          | 3/6           | <b>6/6</b>            |

**Why this table is the scalability answer**

ANPR is not a flat per-camera cost. It is a **policy choice about which cameras carry analytics**, backed by measurement rather than assertion.

| Tier                    | Share  | Workload                                   |
|-------------------------|--------|--------------------------------------------|
| Tier 1: Continuous ANPR | 5-10%  | Highways, border posts, ANPR-sited cameras |
| Tier 2: Event-triggered | 20-30% | Urban junctions                            |
| Tier 3: Registry & view | 60-75% | Coverage without analytics cost            |

**Speaker note:** Quoting *both* modes, and the zero, is what shows the work. If asked why the live grid yields nothing, go straight to Slide 10: we measured it stage by stage and the answer is camera siting, not the software.

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## Slide 7: Correlation, Alerting & Route Reconstruction

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### Watchlist Matching: Exact, then Fuzzy

Every detection is checked before its transaction commits: exact match, then `pg_trgm` trigram similarity > 0.7. A match writes an alert and broadcasts it over WebSocket with the watchlist's own reason and severity.

```
SELECT *, similarity(plate_text, :q) AS score
FROM detections WHERE plate_text % :q
ORDER BY score DESC, detected_at;
```

**Fuzzy search is not a nicety.** ANPR will misread a character, and exact-match-only search fails live on a plate you genuinely detected. Verified: searching `GJ01AB1234` returns the sightings stored as `GJ01AB1234` .

### Route reconstruction

Sightings ordered by time → great-circle distance and elapsed time between consecutive pairs → implied speed → **transitions above ~150 km/h flagged and shown** (not silently dropped).

A flagged transition tells an investigator either that a plate was misread or that two vehicles share a similar registration. Both are useful. A system that hides its own uncertainty is harder to trust than one that surfaces it.

**Speaker note:** Demonstrate the fuzzy search live if there is time. It is the single most persuasive twenty seconds in the demo.

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## Slide 8: Security, Privacy & Accountability

| Control            | Implementation                                                                                           |
|--------------------|----------------------------------------------------------------------------------------------------------|
| Access control     | JWT: three roles <b>enforced per route</b> , not just documented                                         |
| <b>Audit trail</b> | Actor from the <b>verified token</b> , never a request parameter, with stated purpose and case reference |
| Data minimisation  | Plate text, timestamp, camera, plate-region crop (not continuous video)                                  |
| Residency          | All inference local: no frame or crop leaves the deployment                                              |
| Credentials        | Environment only: masked in logs, <b>never serialised to the browser</b>                                 |
| Transport          | CSP, X-Frame-Options DENY, nosniff, HSTS, referrer and permissions policies                              |
| Rate limiting      | 10/min on auth, 300/min elsewhere: video segments exempt                                                 |
| DPDP Act 2023      | Purpose-bound access, role restriction, minimisation, full access trail                                  |

**Verified, not asserted:** a viewer token gets **403** on the audit trail and on plate search, **200** on the camera registry. The eleventh login attempt in a minute returns **429**.

### Closed Government Databases: Handled Correctly

VAHAN, SARTHI, eGujCop and NAFIS have no access route for a hackathon team, and we do not pretend otherwise. We define the request/response contract, implement the adapter, mock the endpoint with realistic records, and document exactly what changes when credentials arrive: base URL, auth, rate limiting, cache, audit hook. Nothing above the adapter changes.

**That is what integration readiness means**, and it survives questioning in a way a claimed live integration would not.

## Slide 9: Scaling to 80,000 Cameras

### We reproduced the 80,000-camera bottleneck at a scale of eight

Opening N concurrent RTSP connections to the sandbox gateway:

| Concurrent | Succeeded | Wall time | Our CPU (20 cores) |
|------------|-----------|-----------|--------------------|
| 2          | 2 / 2     | 12 s      | 4%                 |
| 4          | 4 / 4     | 26 s      | 4%                 |
| 6          | 6 / 6     | 41 s      | 5%                 |
| 8          | 7 / 8     | 88 s      | 4%                 |

Accept times at 8 connections: 4s, 8s, 27s, 33s, 56s, 60s, 63s, 73s (queue buildup).

**Our machine was idle at 4% CPU while the gateway took 73s to accept the eighth connection.** The limit is the single shared ingress, not compute. This is the 80,000-camera bottleneck in miniature.

### Flat central ingestion does not work, and here is the arithmetic

| Quantity                   | Value     |
|----------------------------|-----------|
| Aggregate ingest bandwidth | ≈192 Gbps |
| Central write rate         | ≈24 GB/s  |
| Raw volume, 30 days        | ≈62 PB    |

### Edge-first topology

|                                                   |              |                     |
|---------------------------------------------------|--------------|---------------------|
| camera → DISTRICT EDGE → REGIONAL DC → STATE CORE |              |                     |
| record                                            | aggregate    | registry            |
| transcode                                         | regional     | statewide search    |
| ANPR inference                                    | search       | cross-region routes |
| alert generation                                  | warm storage | dashboards, audit   |
| ↓                                                 | ↓            | ↓                   |
| only metadata,                                    | metadata     | metadata only       |
| alerts, requested                                 | + clips      |                     |
| clips go upstream                                 |              |                     |

Only metadata, alerts, and requested clips traverse the backbone, reducing wide-area bandwidth requirements by **2 to 3 orders of magnitude**.

**Where linearity breaks:** network fan-in at regional aggregation, metadata hot partitions, storage rebuild times, and scatter-gather cross-region route queries. Each needs load-testing before any statewide commitment.

### Why the sandbox ceiling is not our ceiling

| Metric                   | Sandbox Grid                       | Statewide Deployment              |
|--------------------------|------------------------------------|-----------------------------------|
| Cameras per ingest point | 30, one shared gateway             | 50-200 per district edge node     |
| Stream consumption       | Every competing team over internet | Single edge node on local network |
| Across the WAN           | Full video stream to every viewer  | Metadata, alerts, requested clips |

80,000 cameras across ~400 edge nodes is **200 per node**, which is standard server load consistent with our measured 26 concurrent ANPR streams per machine.

**Speaker notes:** If asked "your wall struggles at 9 cameras, how will you do 80,000?", go straight to the 4% CPU figure. The answer: *the sandbox is one shared gateway serving every team; no production deployment looks like that, and our measurements prove the compute side has enormous headroom while shared ingress fails first. This is precisely why the architecture is edge-first.*

Then: "A measured benchmark of 26 streams with a transparent projection is worth more than an unsupported claim of 80,000."

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## Slide 10: Platform Capabilities & Transparent Evaluation

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### Working, demonstrated, reproducible

✓ 30 cameras onboarded with location provenance ✓ GIS map with department, status and confidence layers ✓ Live video wall: HLS grid plus WebRTC hero tile ✓ Continuous ANPR pipeline with tiled inference, track voting and plate grammar ✓ Exact, partial and fuzzy plate search ✓ Automatic watchlist matching and real-time alerts ✓ Route reconstruction with impossible-transition flagging ✓ Output report: XLSX and PDF formats ✓ Audit trail with purpose binding ✓ 57 automated tests | 6/6 ground-truth plate accuracy, 0 false positives

### Honest limitations

- **The pipeline reads no valid plates from this grid, and we know exactly why.** All 30 cameras scouted. 29 are wide-area PTZ overviews yielding only roadside signage, correctly rejected. **cam12, a toll plaza, is the exception:** 25 detections of one real truck plate in 100 seconds, 8 of 10 characters recovered, failing only on 5 px per character. Seven preprocessing variants change nothing. **The limit is optics, not software:** the recommendation is specific to deploy targeted cameras at toll plazas and checkpoints first (evidence in [DOCS/evidence/](#) ).
- Camera coordinates are **geocoded, not surveyed:** accurate to the site, not the pole.
- Department attribution is **inferred** from site type and labelled as such.
- The route is a **sequence of point sightings:** the road-snapped line is an interpolation, returned separately and never presented as evidence.
- Single-node prototype without HA/DR: scale-out path is fully designed and documented in HLD Section 9.

**Speaker note:** Do not rush this slide. NFSU and DA-IICT reviewers will probe; having already named the weaknesses converts a hostile question into agreement.

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## Slide 11: Impact on Policing & Strategic Roadmap

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**Today:** An investigator with a registration number calls each department, waits for manual video scrubbing, and assembles a timeline by hand over several days.

**With Sentinel:** The registration number is typed once. Every sighting across every camera and timestamp appears on an interactive map within seconds because indexing is continuous.

| Capability                 | Operational value                                           |
|----------------------------|-------------------------------------------------------------|
| Continuous indexing        | The route exists before it is asked for                     |
| Automatic watchlist alerts | A stolen vehicle announces itself; nobody watches a wall    |
| Fuzzy search               | A single misread character does not lose the vehicle        |
| Cross-department registry  | One question, one answer, regardless of who owns the camera |
| Purpose-bound audit        | Surveillance capability with accountability attached        |

**Scope:** Models 1 and 2 (chosen strategically rather than defaulted to). Models 2, 3 and 4 are competing answers to departmental fragmentation: Model 2 connects directly and optimizes for the operator, Model 3 federates, Model 4 consolidates. Model 1 is foundational and pairs with exactly one of them. At 30 cameras under one operator, a federation layer would be ceremony around a problem this deployment does not have. Model 3's deliverable also requires federating two *different* VMS platforms and we have access to one; Model 4's requires an 80,000-camera load test on hardware we do not have. We would rather report one measured finding than four partial builds. Full reasoning: HLD Section 1.3.

**Roadmap:** Federation across departments (Model 3) upon second VMS integration, tiered analytics rollout by district, VAHAN adapter live activation on credential grant, followed by facial recognition on edge nodes.

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## Slide 12: Verification & Reproducibility

| Claim                           | Command / Location                                       |
|---------------------------------|----------------------------------------------------------|
| Throughput, both modes          | <code>python anpr_worker.py --benchmark</code>           |
| 6/6 accuracy, 0 false positives | <code>python tools/make_sample_feed.py --validate</code> |
| Grammar and vision logic        | <code>python -m pytest tests/ -q</code> → 57 passed      |
| Location provenance             | <code>GET /api/v1/ingest/status</code>                   |
| Audit trail                     | <code>GET /api/v1/analytics/audit</code>                 |
| Full API surface                | <code>http://localhost:8000/api/docs</code>              |

**Repository:** <https://github.com/DevamShah1211/Sentinel-Hackathone> **Demo videos:** [DOCS/evidence/Sentinel Video/](#) (also on Google Drive) **Hosted instance:** <http://localhost:8000/> / <http://localhost:5173> (test credentials in submission index)

Every number in this deck is reproduced in `DOCS/MEASUREMENTS.md`, with the exact command that produces it.